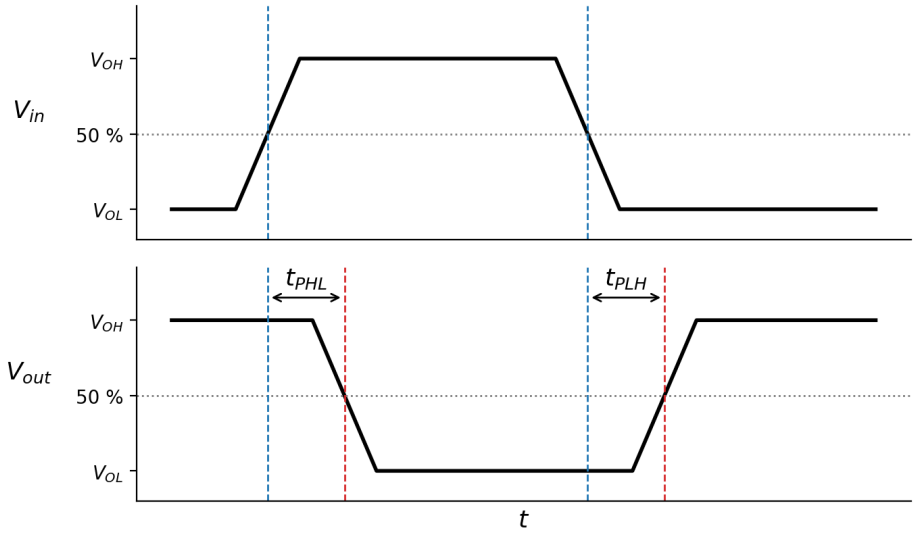
Exercise 02 – Solution Summary

Task 2.1

Idea: Delays in series add up. One period of a ring = **two** passes around the loop.

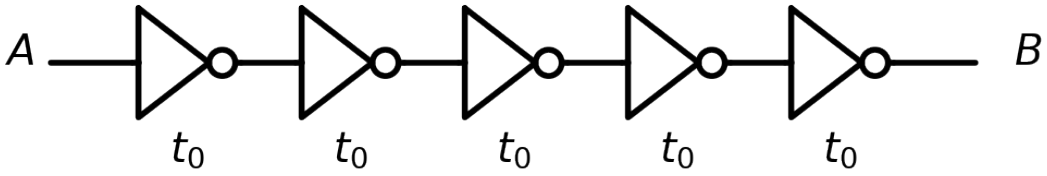
1. **Definition of the inverter delay**



- Measured between **50 % points** of input and output
- Average delay: $t_P = \dfrac{t_{PHL} + t_{PLH}}{2}$
- Symmetrical inverter: $t_{PHL} = t_{PLH} = t_P$

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2. **Task (a):** Delay of $n = 5$ inverters in series from A to B:

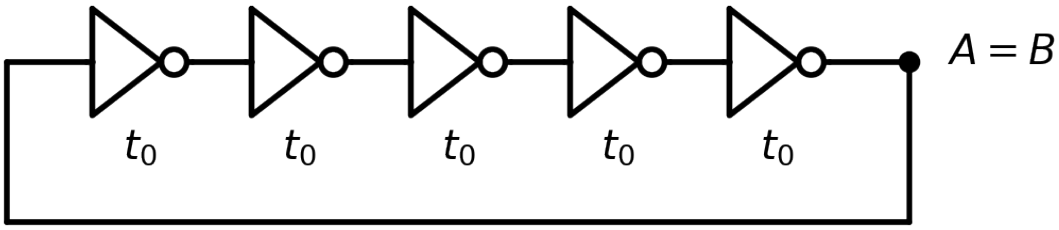


$$t_{AB} = n \cdot t_0 = 5 \cdot 2\,\text{ns} = \boxed{10\,\text{ns}}$$

Since n is odd, $B = \overline{A}$.

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3. **Task (b):** Connecting B back to A gives a ring oscillator:



- Start: $A = 0$
- $+10\,\text{ns}$: A: $0 \to 1$
- $+10\,\text{ns}$: A: $1 \to 0$
- Period = two passes:

$$T = 2 \cdot n \cdot t_0 = 2 \cdot 5 \cdot 2\,\text{ns} = \boxed{20\,\text{ns}}$$

- Frequency:

$$f = \frac{1}{T} = \frac{1}{20\,\text{ns}} = \boxed{50\,\text{MHz}}$$

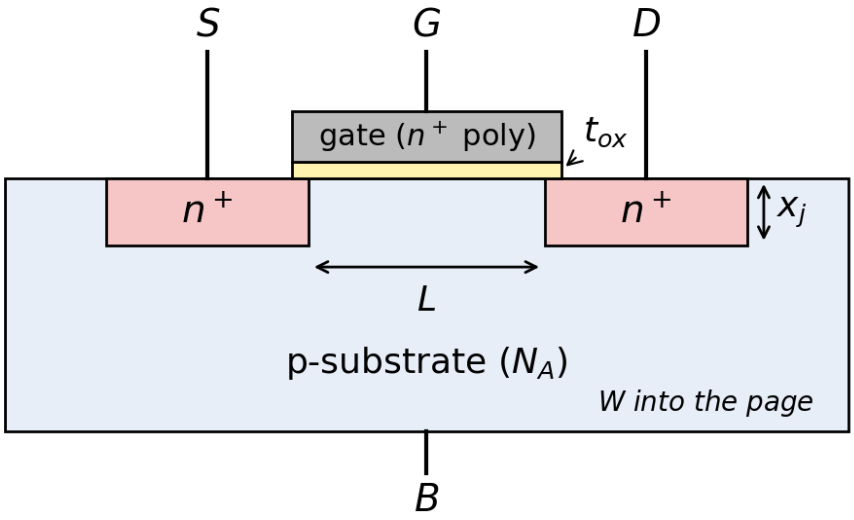
Note:

- Odd n only $\to$ oscillates
- Even $n \to B = A$, bistable element (storage core of a latch)
- Use: measure gate delay, $t_0 = \dfrac{1}{2nf}$
- ps delay $\to$ MHz frequency, easy to measure

Task 2.2

Idea: NMOS on = resistor R_{on}. With R_L it forms a **voltage divider**.

1. **Structure of NMOS**

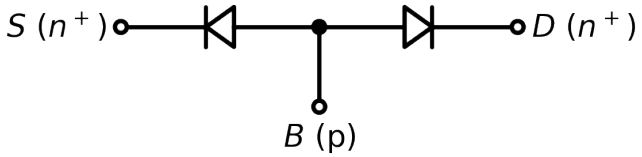


- Two heavily doped n^+ regions (source S and drain D) in a p-doped substrate (bulk B).
- x_j: junction depth, i.e. how deep the n^+ regions reach into the substrate.
- Between them lies the channel with length L and width W, covered by a thin gate oxide of thickness t_{ox} and the poly-silicon gate G.

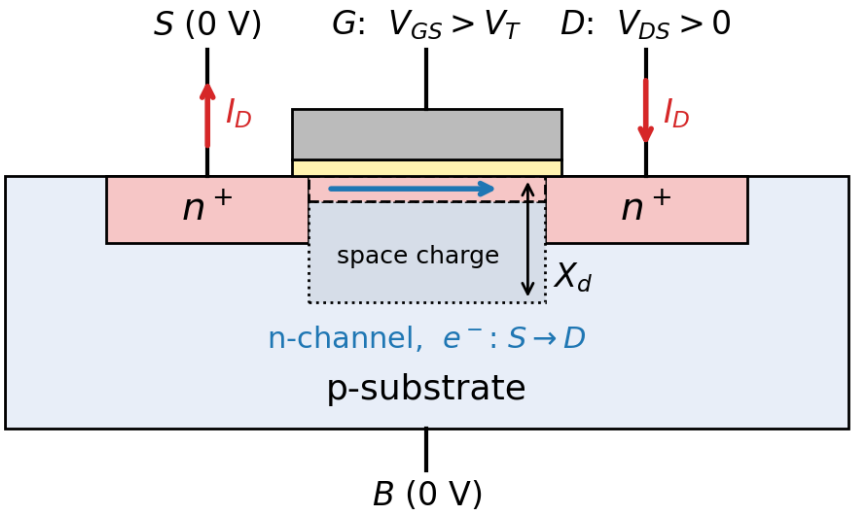
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2. **Operation of NMOS**

- If $V_{GS} < V_T$: n^+/p/n^+ = two back-to-back pn diodes, one is always reverse-biased → **off**, no current.



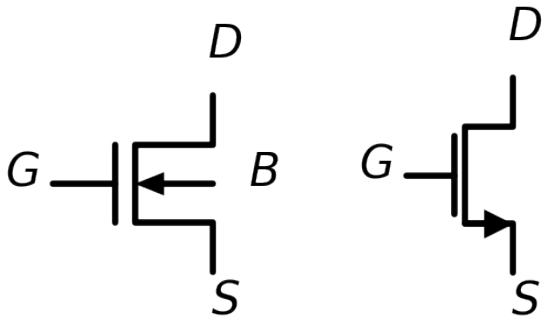
- If $V_{GS} > V_T$: the gate field inverts the p-surface → n-channel (**inversion layer**) connects S and D. Electrons flow $S \rightarrow D$, so the current flows $D \rightarrow S$.



- Below the channel: space charge region (depth X_d)

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3. **Symbol of NMOS**

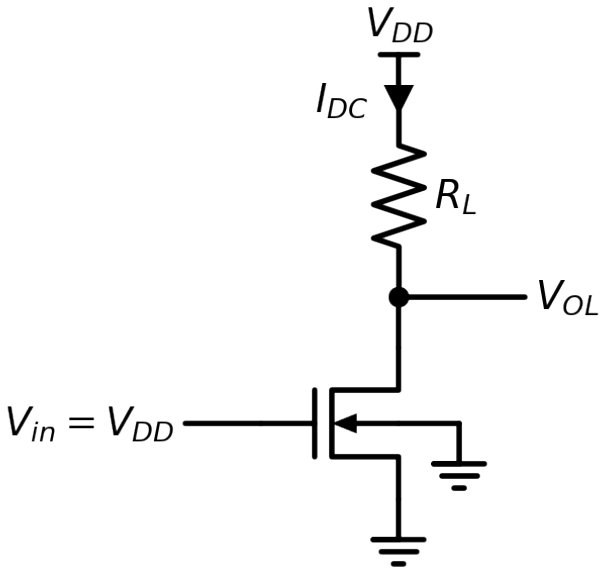


- Left: 4 terminals, bulk arrow = pn junction (p → n)
- Right: simplified, B tied to lowest potential (GND) → omitted; arrow on source = current direction

  Homework: Draw the structure and symbol of a PMOS and analyse its operation.

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4. **Cross current I_{DC} in steady state**



- $V_{in} = V_{DD} \rightarrow$ NMOS on, V_{OL} small $\rightarrow$ linear region $\rightarrow$ resistor R_{on}
- Voltage divider:

$$V_{OL} = V_{DD} \cdot \frac{R_{on}}{R_L + R_{on}}$$

- Same current through both:

$$\boxed{I_{DC} = \frac{V_{DD}}{R_L + R_{on}}}$$

Note:

- Output low $\rightarrow I_{DC}$ flows all the time $\rightarrow$ static power $P = V_{DD} \cdot I_{DC}$
- $V_{OL} \approx 0 \rightarrow$ need $R_L \gg R_{on}$
- CMOS fixes both: PMOS replaces R_L and is off when NMOS is on
- Linear: $V_{GS} > V_T$ and $V_{DS} < V_{GS} - V_T$

Task 2.3

Idea: Build V_{T0} term by term (a–d), then shift it to the target by implantation (e).

Constants (all lengths in cm):

- Thermal voltage at room temperature: $kT/q = 0.026$ V
- Elementary charge: $q = 1.6 \cdot 10^{-19}$ As
- Intrinsic carrier concentration of silicon: $n_i = 1.45 \cdot 10^{10}$ cm^{-3}
- Permittivities: $\epsilon_0 = 8.85 \cdot 10^{-14} \frac{\text{As}}{\text{Vcm}}$, $\epsilon_{si} = 11.7\,\epsilon_0 \approx 1.0 \cdot 10^{-12} \frac{\text{As}}{\text{Vcm}}$, $\epsilon_{ox} \approx 4\,\epsilon_0$
- Band gap of silicon: $E_g = 1.1$ eV, the intrinsic level E_i lies in the middle (0.55 eV from each band edge).

Given: $N_A = 3 \cdot 10^{17}\,\text{cm}^{-3}$, $t_{ox} = 22\,\text{Å} = 22 \cdot 10^{-8}\,\text{cm}$, $N_D(\text{gate}) = 10^{20}\,\text{cm}^{-3}$, $N_{ox} = 2 \cdot 10^{10}\,\text{cm}^{-2}$.

1. **Task (a): Band bending for strong inversion, Φ_F and $2|\Phi_F|$**

   - The Fermi potential Φ_F is the distance between the Fermi level and the intrinsic level, $q\Phi_F = E_F - E_i$. It is < 0 for p-type and > 0 for n-type substrate:

$$\Phi_{Fp} = \frac{kT}{q} \ln \frac{n_i}{N_A} = 0.026\,\text{V} \cdot \ln \frac{1.45 \cdot 10^{10}\,\text{cm}^{-3}}{3 \cdot 10^{17}\,\text{cm}^{-3}} \approx -0.44\,\text{V}$$

   - **Strong inversion** is reached when the electron concentration at the surface equals the hole concentration in the substrate, i.e. the surface potential is $\Phi_s = -\Phi_{Fp}$. The band bending relative to flat-band is:

$$|\Phi_s - \Phi_{Fp}| = 2|\Phi_{Fp}| = 2 \cdot 0.44\,\text{V} = \boxed{0.88\,\text{V}}$$

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2. **Task (b): Space charge region, width X_d and charge Q_{B0}**

   - Maximum depletion width X_d:

$$X_d = \sqrt{\frac{2\epsilon_{si}\,|\Phi_s - \Phi_F|}{qN_A}} = \sqrt{\frac{2 \cdot 11.7 \cdot 8.85 \cdot 10^{-14}\,\frac{\text{As}}{\text{Vcm}} \cdot 0.88\,\text{V}}{1.6 \cdot 10^{-19}\,\text{As} \cdot 3 \cdot 10^{17}\,\text{cm}^{-3}}} \approx 6 \cdot 10^{-6}\,\text{cm} = \boxed{60\,\text{nm}}$$

   - Fixed negative charge (ionized acceptors) per area:

$$\begin{aligned} Q_{B0} &= -\sqrt{2qN_A\epsilon_{si}\,|-2\Phi_F|} \\ &= -\sqrt{2 \cdot 1.6 \cdot 10^{-19}\,\text{As} \cdot 3 \cdot 10^{17}\,\text{cm}^{-3} \cdot 1.0 \cdot 10^{-12}\,\frac{\text{As}}{\text{Vcm}} \cdot 0.88\,\text{V}} \\ &\approx \boxed{-3 \cdot 10^{-7}\,\text{C/cm}^2} \end{aligned}$$

   Note: Consistency check: $Q_{B0} = -qN_AX_d = -1.6 \cdot 10^{-19}\,\text{As} \cdot 3 \cdot 10^{17}\,\text{cm}^{-3} \cdot 6 \cdot 10^{-6}\,\text{cm} \approx -3 \cdot 10^{-7}\,\text{C/cm}^2$.

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3. **Task (c): Oxide capacitance C_{ox} and body factor γ**

   - Oxide capacitance per area:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{4 \cdot 8.85 \cdot 10^{-14}\,\frac{\text{As}}{\text{Vcm}}}{22 \cdot 10^{-8}\,\text{cm}} \approx \boxed{1.6 \cdot 10^{-6}\,\text{F/cm}^2}$$

   - Body factor:

$$\gamma = \frac{\sqrt{2q\epsilon_{si}N_A}}{C_{ox}} = \frac{\sqrt{2 \cdot 1.6 \cdot 10^{-19}\,\text{As} \cdot 11.7 \cdot 8.85 \cdot 10^{-14}\,\frac{\text{As}}{\text{Vcm}} \cdot 3 \cdot 10^{17}\,\text{cm}^{-3}}}{1.6 \cdot 10^{-6}\,\text{F/cm}^2} \approx \boxed{0.2\,\text{V}^{1/2}}$$

   Note: γ describes how much V_T increases with a source-bulk voltage: $V_T = V_{T0} + \gamma\left(\sqrt{|-2\Phi_F + V_{SB}|} - \sqrt{|-2\Phi_F|}\right)$. That is why $V_{SB} = 0$ is given in Task (d).

4. **Task (d): Threshold voltage V_{T0} ($V_{SB} = 0$)**
   - The threshold voltage consists of four terms:

$$V_{T0} = \Phi_{GC} - 2\Phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

   - **Work function difference** gate–channel Φ_{GC} (Φ = work function, ϕ = Fermi potential):

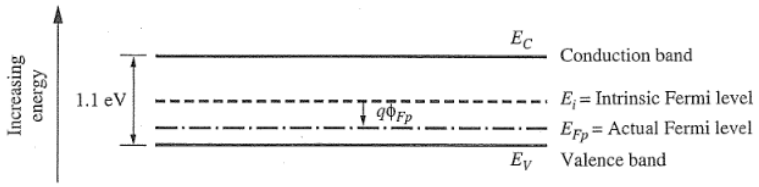


Figure 2.4
Energy-band diagram for doped *p*-type silicon.

$q\phi_{Fp}$ = distance $E_i \to E_{Fp}$; $E_g = 1.1$ eV, E_i in the middle $\to E_C - E_i = 0.55$ eV

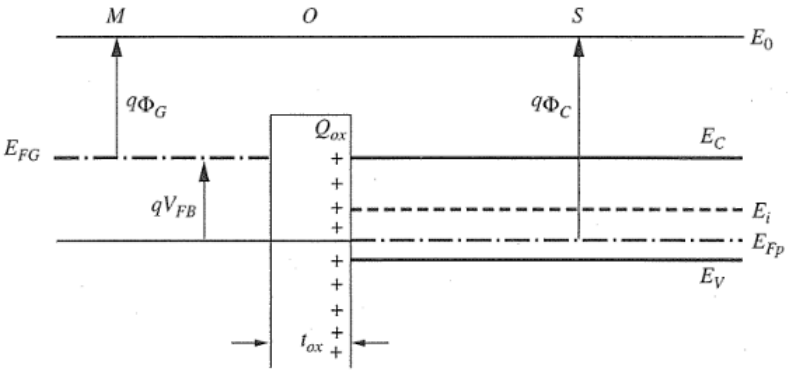


Figure 2.6
Flat-band condition in the MOS system.

   - Work functions from the figure (E_0 = vacuum level):

$$q\Phi_G = E_0 - E_{FG}, \qquad q\Phi_C = E_0 - E_{Fp}$$

$$\Rightarrow \quad q\Phi_{GC} = q\Phi_G - q\Phi_C = E_{Fp} - E_{FG}$$

   - Gate is n^+ poly-Si → same E_i as substrate. With $q\phi_F = E_{Fp} - E_i$:

$$E_{Fp} = E_i + q\phi_{Fp}, \quad \phi_{Fp} = -0.44\,\text{V}$$

$$E_{FG} = E_i + q\phi_G, \quad \phi_G = +0.55\,\text{V} \;\; (N_D = 10^{20}\,\text{cm}^{-3} \to E_{FG} \approx E_C)$$

   - Insert:

$$\Phi_{GC} = \Phi_G - \Phi_C = \phi_{Fp} - \phi_G = -0.44\,\text{V} - 0.55\,\text{V} = -0.99\,\text{V}$$

- **Depletion charge** (from Tasks (b) and (c)):

$$\frac{Q_{B0}}{C_{ox}} = \frac{-3 \cdot 10^{-7}\,\text{C/cm}^2}{1.6 \cdot 10^{-6}\,\text{F/cm}^2} \approx -0.188\,\text{V}$$

- **Oxide charge**:

$$\frac{Q_{ox}}{C_{ox}} = \frac{qN_{ox}}{C_{ox}} = \frac{1.6 \cdot 10^{-19}\,\text{As} \cdot 2 \cdot 10^{10}\,\text{cm}^{-2}}{1.6 \cdot 10^{-6}\,\text{F/cm}^2} = 0.002\,\text{V}$$

- Insert:

$$V_{T0} = -0.99\,\text{V} - (-0.88\,\text{V}) - (-0.188\,\text{V}) - 0.002\,\text{V} \approx \boxed{0.08\,\text{V}}$$

- **Gate doping:** p^+ gate → $\phi_G = -0.55\,\text{V}$:

$$\Phi_{GC} = -0.44 + 0.55 = 0.11\,\text{V} \;\Rightarrow\; V_{T0} \approx 1.18\,\text{V} \gg 0.4\,\text{V}$$

→ NMOS needs n^+ gate (PMOS: p^+ gate), rest fixed by implantation.

Note: These values come from a simple model and only show the trend; they are not exact values of a real 130 nm device.

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5. **Task (e): Ion implantation dose N_I for $V_T = 0.4\,\text{V}$**

- Implanted acceptors add a negative charge $Q_I = -qN_I$ at the surface (same sign as Q_{B0}):

$$\Delta V_T = -\frac{Q_I}{C_{ox}} = \frac{qN_I}{C_{ox}}$$

- Required shift: $\Delta V_T = 0.4\,\text{V} - 0.08\,\text{V} = 0.32\,\text{V}$

$$N_I = \frac{C_{ox}\,\Delta V_T}{q} = \frac{1.6 \cdot 10^{-6}\,\text{F/cm}^2 \cdot 0.32\,\text{V}}{1.6 \cdot 10^{-19}\,\text{As}} = \boxed{3.2 \cdot 10^{12}\,\text{ions/cm}^2}$$

Note: To **increase** V_T of an NMOS, additional negative charge is needed under the gate, so **acceptors** (e.g. B, boron) are implanted. Donors (e.g. P, phosphorus) would lower V_T.